

One Subject-Level Analysis of Patient fMRI

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Abstract

fMRI has revolutionized the healthcare industry by allowing researchers and medical professionals to observe real-time neural activity. This paper will present an overview of the process of a one-subject-level analysis of fMRI patient data. This approach showcases the importance of preprocessing fMRI data in Neuroimaging and how patient-level analyses can contribute to the broader understanding of neural activity.

Keywords – Functional Magnetic Resonance Imaging (fMRI); Statistical Parametric Mapping (SPM); General Linear Model (GLM); Signal to Noise Ratio (SNR);

Introduction

fMRI is a technique derived from magnetic resonance imaging that focuses on mapping out time-varying changes in the brain’s neural activity. These changes are consequent to task-induced cognitive states such as “move” or “rest”. Since 1990, fMRI has used various applications such as surgical planning, monitoring treatment outcomes, and pharmacology biomarking. [1]. However, several methodological challenges arise during the fMRI acquisition stage. fMRI is subject to various types of “noise” which include subject motion and unwanted signal acquisition (anatomical, environment, thermal, baseline drift, etc.). To mitigate these effects, essential preprocessing will be utilized. This method includes realignment, coregistration, and smoothing. Finally, the GLM is used to model task-related neural activity [2]. By utilizing these steps in fMRI, researchers can build upon high-quality fMRI data that be utilized in further clinical applications and contribute to an expanding understanding of brain functionality.

Methodology

A patient sample data from a 3T scan was collected and downloaded from the OpenNeuro data set. The patient data included the anatomical data files and functional MRI patient data files. [3]. The SPM (Statistical Parametric Mapping) Matlab module was downloaded and implemented into the environment as a method to visualize fMRI images and manipulate the layout. [4]. The Tapas Matlab module was also downloaded to do preprocessing for the fMRI data in Matlab script. This module includes the functionality

to easily visualize numerical image data, volumetric image data, quality checking, image realignment, coregistration, smoothing, and the General Linear Model. [5]. When initially visualizing the image, the functional image is not clearly defined as the anatomical image due to its lower resolution. This is due to the T1-weighted image (anatomical) having a very slow collection rate in comparison to the functional images’ fast collection rate. [6]. This results in noise in the functional images such as image misalignments, timing errors, spatial variations, and a lower SNR.

The fMRI images must be pre-processed before visualizing them on the SPM platform. First, the time-variant functional images are realigned to the first indexed image. This corrects the alignment of the 3-D spatial coordinates to a reference image. Figure 1 shows the realignment parameters generated from the time-variant fMRI acquisition. Next, the functional fMRI image set is coregistered to anatomical images. This warps the functional image dataset into a standardized space such that it can easily analyzed. Then, the functional image dataset is smoothed by averaging the cut-off over nearby voxels. This reduces some of the noise and enhances the overall signal strength. Finally, the GLM was used to model a “linear fit” to the pre-processed data to assist in understanding how predictors in the model vary the data. The basic model is defined as:

$$Y = X\beta + \varepsilon$$

Where Y is the fully preprocessed fMRI data, X is a matrix of predictors, β is the weights of the predictors and ε is the residual errors. The predictors used in the GLM are the timing, durations, and onsets of the fMRI data. The residual error is set to the realignment parameter dataset generated after realigning the fMRI dataset together. Within X , only data of the “move” event was segmented into the GLM as the aim of this fMRI image is to capture the observed brain oxygenation activity.

The SPM platform was used to observe the fMRI data. A contrast of [1 0] was applied to the contrast manager to represent the mean activation across the fMRI data. No masking or threshold extension was applied and a threshold p value of 1×10^{-3} was applied. An overlay of the incongruent-congruent contrast image is generated to visualize the voxel clusters that pass the threshold p-value, as seen in Figure 2.

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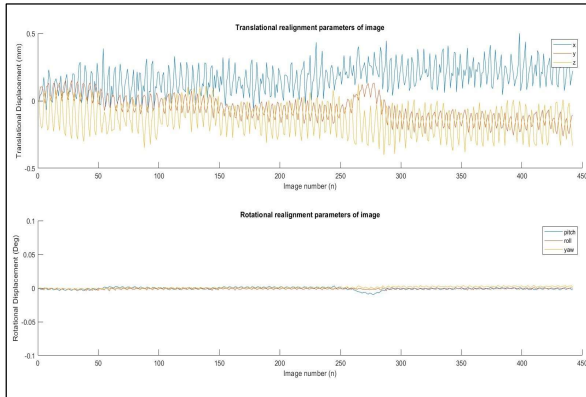


Figure 1 – Time variant image realignment

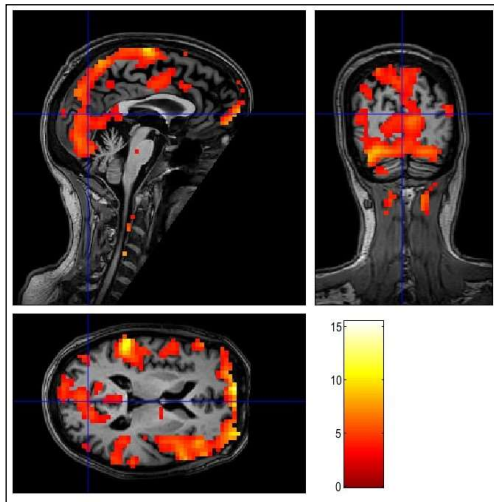


Figure 2 - Incongruent-congruent contrast overlay

Results and Discussion

Figure 1 highlights the importance of preprocessing the data. During the fMRI image acquisition, the time-variant realignment graph shows a spike in the Y-axis and the pitch data. This indicates that the patient physiologically moved their head upward during the fMRI scan. Additionally, the number of offset translations present in Figure 1 shows indicates that the translational data acquisition was not synchronized to the same origin as the reference image. This could either be due to limitations in the spatial hardware, noise present in the environment or the patient subconsciously moving during the duration of the scan. If this data were to be processed without the realigning, then the fMRI data would feature movement artifacts that would translate into a compromise of the quality and validity of the neural activity. Figure 2 reveals the processed model and overlay of the incongruent-congruent contrast of the patient fMRI data. As indicated in the figure, the high correlation of neural activity is visualized by the bright colouring and a lower correlation by the darker red

colouring. When “move” data was segmented out, the passive neural activity was filtered out, resulting in only key areas of the brain’s cortex being visualized. When the threshold p-value is increased, the level of low-correlation voxels is reduced, leaving only high-correlated neural activity voxels to be displayed.

When running the Matlab script, the realignment, coregistration, smoothing and GLM took a considerable amount of time to process. A major improvement that should be considered is run time efficiency. When experimenting with the platform, modifications to the resolution parameters was implemented. The resulting data only reduced the processing time by a small fraction however, it drastically reduced image quality severely affected the Incongruent-congruent contrast overlay.

Conclusions

The single-subject fMRI analysis presented in this paper demonstrates the process of raw fMRI data into the visual mapping of the brain functionality. This paper also highlights the importance of preprocessing data for clinical applications and future fMRI research.

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Biography

Tevyn Vergara is a fifth-year Electrical & Biomedical Engineering student at UQ who is aiming to graduate in Semester 2, 2025. He is passionate about medical technology and has worked on multiple medical engineering projects, including ECG, Physiological Modelling Simulations, and MRI.